

Running Regression Analysis for Beta Estimation

HDFC Bank vs NIFTY 50 | Tutorial 3

Tutorial 3: Beta Estimation & Interpretation

Using regression analysis, estimate the beta of HDFC Bank with respect to the NIFTY 50 index over the past one year. Interpret the results in terms of systematic vs unsystematic risk. Discuss whether the stock is suitable for conservative investors.

Understanding Beta, Alpha & R-Square (A Simple Guide Using Stock vs Index Daily Returns Regression)

What Are We Measuring?

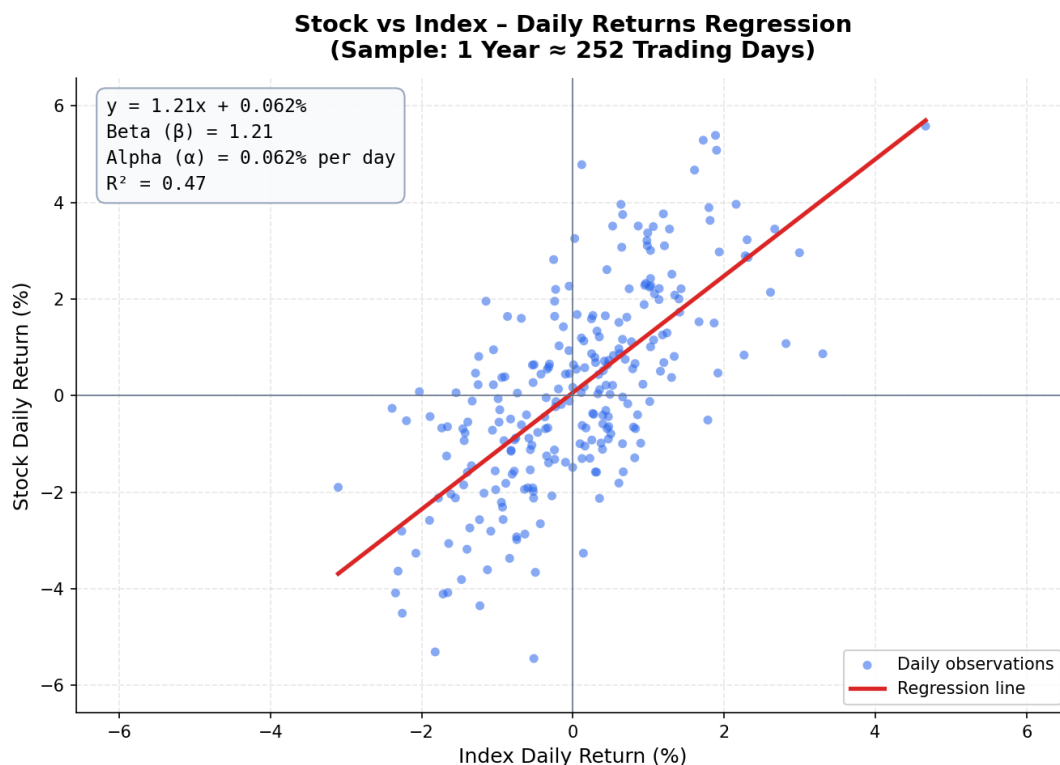
When you own a stock, part of its daily ups and downs comes from the overall market (the index), and part comes from the company itself. A simple linear regression helps us separate these two effects and answer three practical questions:

- How sensitive is the stock to the market? → **Beta**
- Did the stock outperform or underperform after adjusting for risk? → **Alpha**
- How much of the stock's movement is explained by the market? → **R-Square**

We plot the stock's daily returns against the index's daily returns and draw the best-fitting straight line. That line and the statistics around it give us Beta, Alpha and R².

Sample Scatter Chart with Regression Line

Below is a realistic example using one year of daily returns (≈ 252 trading days). Each blue dot is one day's pair of returns. The red line is the fitted regression line.



From the sample chart the equation is approximately:

$$\text{Stock Return} = 1.21 \times \text{Index Return} + 0.062\%$$

Metric	Value (Sample)	What it means in plain English
Beta (β)	≈ 1.21	Stock moves about 21% more than the index on average
Alpha (α)	$\approx +0.062\%$ per day	Small positive daily outperformance ($\approx +15.6\%$ annualized)
R-Square (R^2)	≈ 0.47	About 47% of the stock's moves are explained by the index

1. Beta (β) – The Slope of the Line

Beta measures how sensitive the stock is to movements in the index. It is literally the slope of the red line in the chart. A steeper line means higher beta.

Beta Value	Simple Meaning	Practical Implication
$\beta = 1.0$	Moves exactly with the market	Same systematic risk as the index
$\beta > 1.0$ (e.g. 1.21)	Amplifies market moves	More volatile $\rightarrow$ higher expected long-term return
$\beta < 1.0$	Dampens market moves	Less volatile $\rightarrow$ lower market risk
$\beta \approx 0$	Almost no link to the market	Mostly company-specific risk
β negative	Moves opposite to the market	Rare for ordinary stocks; can act as a hedge

In the sample: Beta ≈ 1.21 means that when the index rises 1% in a day, the stock is expected to rise about 1.21% on average. When the index falls 1%, the stock is expected to fall about 1.21%. This is called systematic risk — the risk that cannot be diversified away.

2. Alpha (α) – The Intercept of the Line

Alpha is the point where the regression line crosses the vertical (Y) axis. It is the average return the stock delivered that is not explained by the index's movement.

- Positive alpha $\rightarrow$ the stock outperformed what its beta (risk level) would predict.
- Negative alpha $\rightarrow$ the stock underperformed relative to its risk.
- Alpha near zero $\rightarrow$ the stock performed roughly in line with its risk.

In the sample: Alpha $\approx +0.062\%$ per day. This looks tiny, but over a full year it compounds to roughly +15–16% (simple approximation: $0.062\% \times 252$ trading days). In other words, after adjusting for the extra risk the stock took (its higher beta), it still delivered positive excess return during this period.

Important caution: Alpha calculated on only one year of data can easily change sign in the next year. A large positive alpha may reflect temporary luck, sector tailwinds, or genuine skill — we usually need longer periods and more tests before calling it “true skill.”

3. R-Square (R^2) – How Tightly the Points Fit the Line

R^2 tells you what percentage of the stock's daily return variation is explained by the index. The rest is idiosyncratic (company-specific) noise.

R^2 Range	Meaning	What it tells you
0.80 – 0.95	Very strong relationship	Index is an excellent benchmark; beta is reliable
0.50 – 0.80	Decent relationship	Typical for many large-cap stocks
0.20 – 0.50	Moderate / weaker link	Stock has significant unique risk (sample ≈ 0.47)
Below 0.20	Weak relationship	Index is a poor benchmark; beta less trustworthy

In the sample: $R^2 \approx 0.47$ means roughly 47% of the stock's daily moves were driven by the overall market. The other 53% came from company-specific news, earnings, management decisions, sector rotation, etc. Because R^2 is moderate, the beta number is useful but not extremely precise — there is still a fair amount of noise.

How to Read the Chart at a Glance

- **Steep red line** → High beta (stock is aggressive).
- **Flatter red line** → Low beta (stock is defensive).
- **Line crosses Y-axis well above zero** → Positive alpha.
- **Line crosses Y-axis below zero** → Negative alpha.
- **Dots tightly clustered around the line** → High R^2 (clean relationship).
- **Dots widely scattered** → Low R^2 (lots of idiosyncratic risk).

Summary

In the example chart the stock has a beta of about 1.21, a small positive daily alpha, and an R^2 of roughly 0.47. In simple words:

“This stock tends to rise and fall a bit more than the market. Over the past year it delivered some extra return beyond what its higher risk level would suggest. However, only about half of its daily movements were explained by the market — the rest came from company-specific factors.”

Practical Tips When You Run This Yourself

- Use the same dates for both the stock and the index (no missing days).
- Returns should be in decimal or percent form — just be consistent.
- One year of daily data is a common starting point, but betas and alphas can change over time.
- Always look at the scatter plot — numbers alone can hide outliers or non-linear patterns.

Steps to perform:

1. Objective

You have one year of daily returns for a stock and a market index. The goal is to run a simple linear regression to estimate:

- **Beta (β)** – how sensitive the stock is to the index (the slope)
- **Alpha (α)** – the average excess return not explained by the index (the intercept)
- **R-Square (R^2)** – how much of the stock’s movement is explained by the index

The underlying model is:

$$\text{Stock Return} = \alpha + \beta \times \text{Index Return} + \text{error}$$

2. Prepare Your Data

Arrange the data in two columns (same number of rows, same dates):

Row	Column A (optional)	Column B – Stock	Column C – Index
1 (header)	Date	Stock Return	Index Return
2	2025-08-21	0.0125 (or 1.25%)	0.0080 (or 0.80%)
3 ... 253	...	...	...

Important checks before you start:

- Both series must have the same length and matching dates (no missing days).
- Use either decimals (0.0125) or percentages (1.25) — be consistent for both series.
- Approximately 252 rows of data for one trading year is ideal.

3. Method 1 – Quick Formulas (Fastest)

These three formulas give you the answers instantly and update automatically if the data changes. Assume stock returns are in B2:B253 and index returns are in C2:C253.

Statistic	Excel Formula	What you get
Beta (β)	=SLOPE(B2:B253,C2:C253)	Slope of the line
Alpha (α)	=INTERCEPT(B2:B253,C2:C253)	Intercept (daily)
R-Square (R^2)	=RSQ(B2:B253,C2:C253)	Goodness of fit (0 to 1)

Note: In SLOPE / INTERCEPT / RSQ the first argument is always the Y (stock) range and the second is the X (index) range.

4. Method 3 – LINEST (Advanced Array Formula)

LINEST returns multiple regression statistics at once. In modern Excel 365 you can simply type it and press Enter (it spills automatically).

Formula:

=LINEST(B2:B253,C2:C253,TRUE,TRUE)

It returns a block of values. The most important ones are:

Position	Meaning
Row 1, Column 1	Beta (slope)
Row 1, Column 2	Alpha (intercept)
Row 3, Column 1	R^2

5. Create the Scatter Chart & Show the Line

A picture is the best way to understand the relationship. Follow these steps:

Step 1 – Insert the scatter plot

1. Select the Index returns column and the Stock returns column together (or select the whole data block).
2. Go to Insert → Charts → Scatter → choose the plain Scatter (markers only).
3. Excel plots Index on the X-axis and Stock on the Y-axis.

Step 2 – Add the regression (trend) line

1. Click the chart to select it.
2. Click the + (Chart Elements) button on the top-right of the chart.
3. Tick Trendline. Then click the small arrow next to Trendline → More Options...
4. In the Format Trendline pane make sure Linear is selected.
5. Tick “Display Equation on chart”.
6. Tick “Display R-squared value on chart”.

Step 3 – Make the chart readable

1. Add chart title (e.g. “Stock vs Index – Daily Returns Regression”).
2. Add axis titles: X = “Index Daily Return”, Y = “Stock Daily Return”.
3. Optionally format the equation box (larger font, move to a clear spot).

Step 4 – Interpret Results